

HW 7: 7.17, 7.38, 8.25, 8.66

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7.17) Figure (a) results in turbulent over entire surface. Therefore, lowest heat transfer is associated with fig (1).

$$T_{\text{mean}} = \frac{35 + 10}{2} = 22.5^\circ\text{C} \quad T_{\text{mean air props}}: k = 0.0253 \frac{\text{W}}{\text{m}\cdot\text{K}}, \nu = 1.539 \times 10^{-6} \text{ m}^2/\text{s}$$

→ in fig (1), transition will occur.

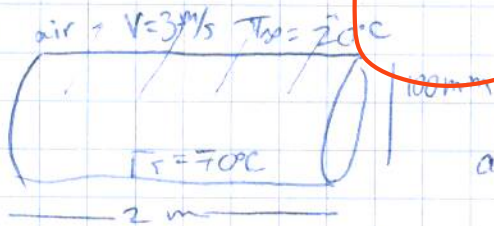
$$Re_c = 5 \times 10^5 = \frac{Vx}{\nu} \Rightarrow 5 \times 10^5 = \frac{2.5 \times k}{1.539 \times 10^{-6}} \quad V = 0.03 \text{ m}$$

$$Nu_L = (0.037 \times (1.6 \times 10^7)^{1/4} - 871) \cdot 0.73^{0.33} = 18552$$

$$\text{fig (2)} \quad Nu_{L2} = 0.037 (1.67 \times 10^7)^{1/4} \cdot (0.73)^{0.33} = 19337 \quad Nu_L = h_L \cdot L = 18552$$

$$h_{L1} = 469.7 \text{ W/m}^2\cdot\text{K} \quad q_1 = h_{L1} A (T_s - T_\infty) = 469.7 \times 0.5 \times 25 = 5871 \text{ W}$$

7.38.)



$$T_f = \frac{T_s + T_\infty}{2} = \frac{70 + 20}{2} = 45^\circ\text{C}$$

$$\text{air @ } 45^\circ\text{C}: \rho = 1.189 \frac{\text{kg}}{\text{m}^3}, k = 0.02699 \frac{\text{W}}{\text{m}\cdot\text{K}}, \mu = 1.941 \times 10^{-5} \text{ kg/m}\cdot\text{s}, \nu = 1.75 \times 10^{-5} \text{ m}^2/\text{s}, Pr = 0.7241$$

$$Re = \frac{VD}{\nu} = \frac{3 \cdot 0.1}{1.75 \times 10^{-5}} = 17142.857$$

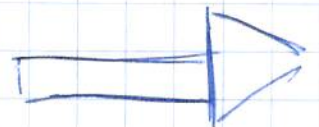
Nusselt no. from Churchill-Bern for crossflow.

$$Nu = 0.3 + \frac{0.62 Re^{1/2} Pr^{1/3}}{[1 + (\frac{0.4}{Pr})^{1/3}]^{1/4} [1 + (\frac{Re}{28200})^{5/8}]^{4/5}} = 73.16$$

$$\text{convective heat transfer coeff: } h = \frac{k}{D} \cdot Nu = \frac{0.02699}{0.1} \times 73.1562 = 19.745 \frac{\text{W}}{\text{m}^2\cdot\text{K}}$$

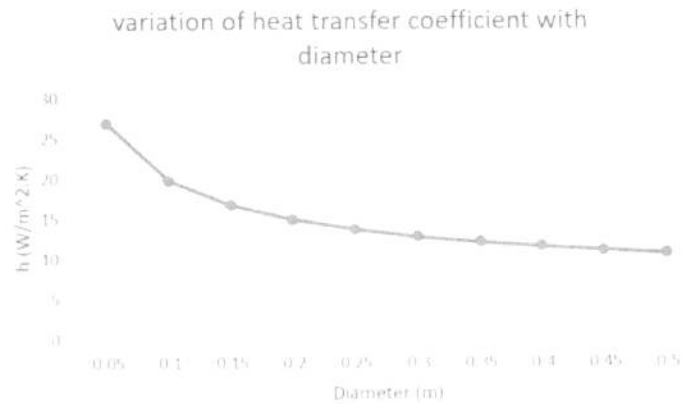
$$\text{Surt Area: } A_s = \pi D L = \pi \cdot 0.1 \times 2 = 0.6283 \text{ m}^2$$

$$\text{convective thermal resist. } R_{th} = \frac{1}{h A_s} = \frac{1}{19.745 \cdot 0.6283} = 0.0806 \frac{\text{K}}{\text{W}}$$



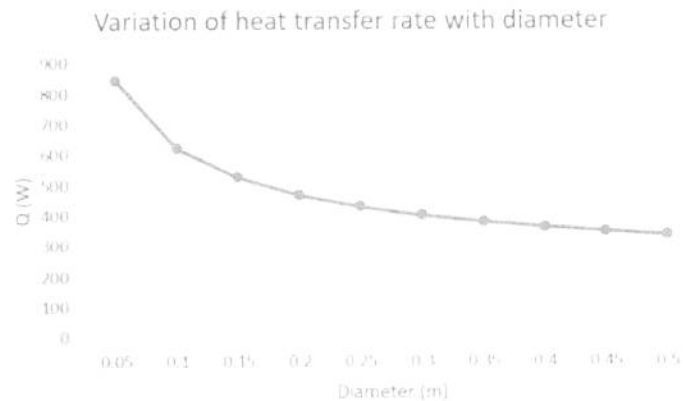
S.No.	D (m)	Re	Nu	h (W/m ² ·K)	Q (W)
1	0.05	8571.428571	49.66089949	26.80695355	842.1404456
2	0.1	17142.85714	73.15621135	19.74486144	620.2848222
3	0.15	25714.28571	92.56502609	16.65553369	523.233591
4	0.2	34285.71429	109.8983102	14.83077696	465.9088581
5	0.25	42857.14286	125.9251575	13.59488001	427.0831554
6	0.3	51428.57143	141.0359308	12.68853257	398.6102507
7	0.35	60000	155.4604278	11.98821985	376.6099266
8	0.4	68571.42857	169.3471559	11.42669934	358.9697598
9	0.45	77142.85714	182.7984985	10.96384772	344.4292761
10	0.5	85714.28571	195.8884846	10.5740604	332.1841074

Consider the following plot between the diameter and the convective heat transfer coefficient:



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Consider the following plot between the heat transfer rate and the diameter:



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8.25) $\frac{320-310}{320-310} = \frac{14.1 \times 10^{-2} - \mu_i}{14.1 \times 10^{-2} - 25.3 \times 10^{-2}} \rightarrow 16.3 \times 10^{-2} \text{ N.s/m}^2$ $\frac{360-353}{360-350} = \frac{2.52 \times 10^{-2} - \mu_o}{2.52 \times 10^{-2} - 3.56 \times 10^{-2}}$
 inlet $\mu_o = 3.25 \times 10^{-2} \text{ N.s/m}^2$ outlet

$Re_{in} = \frac{4\dot{m}}{\pi D \mu_i} = \frac{4 \times 1}{\pi \cdot 0.005 \cdot 16.3 \cdot 10^{-2}} = 1562.3$ (laminar) $Re_{out} = \frac{4}{0.005 \pi \cdot 3.25 \cdot 10^{-2}} = 7835.3$ (turbulent)

Re when flow laminar to turb. $Re = 2300 = \frac{4\dot{m}}{\pi D \mu} \rightarrow \mu = \frac{4\dot{m}}{2300 \pi D} = \frac{4}{0.005 \pi \cdot 2300} = 1.1 \times 10^{-2} \text{ Pa.s}$

$\bar{T}_{m1} = \frac{45+52}{2} = 48.5^\circ\text{C} = (48.5+273)\text{K} = 321.5\text{K}$ @ 321.5K

$C_p = 1999 \text{ J/kg.K}$, $\mu_i = 13.2 \times 10^{-2} \text{ N.s/m}^2$, $k_i = 0.143 \text{ W/m.K}$, $Pr_i = 1850$

$Re_{D1} = \frac{4\dot{m}}{\pi D \mu_i} = 1929.15$ hydrodynamic entry length: $X_{fd,h} = 0.05 Re_{D1}$, $D = 0.05 \cdot 1929.15 \cdot 0.005 = 0.48 \text{ m}$

Thermal entry length $X_{fd,t} = X_{fd,h} Pr_i = 0.48 \cdot 1851 = 890$ $\frac{T_s - T_{m,t}}{T_s - T_i} = \exp\left(\frac{-\pi D L_1 \bar{h}_1}{\dot{m} C_{p,i}}\right)$

$\ln\left(\frac{T_s - T_{m,t}}{T_s - T_i}\right) = \frac{-\pi D L_1 \bar{h}_1}{\dot{m} C_{p,i}}$ $\bar{h}_1 L_1 = \frac{-\dot{m} C_{p,i}}{\pi D} \ln\left(\frac{T_s - T_{m,t}}{T_s - T_i}\right) = \frac{-1 \cdot 1999}{\pi \cdot 0.005} \ln\left(\frac{150-52}{150-45}\right) = 8780$

$\bar{Nu}_{D1} = \frac{\bar{h}_1 D}{k_i} = 3.66 + \frac{0.0668 \left(\frac{D}{L_1}\right) Re_{D1} Pr_i}{1 + 0.04 \left(\frac{D}{L_1}\right) Re_{D1} Pr_i}^{1/4}$

pipe length @ state 1 $\bar{Nu}_{D1} = 3.66 = \frac{\bar{h}_1 D}{k_i}$ $\bar{h}_1 = \frac{3.66 \cdot 0.143}{5 \text{ mm} \cdot \frac{1}{1000 \text{ mm}}} = 105 \frac{\text{W}}{\text{m}^2\text{K}}$ $L_1 L_1 = 8780$
 $L_1 = 84 \text{ m}$

cross ch. $\bar{Nu}_{D1} = 3.66 + \frac{0.0668 \left(\frac{D}{L_1}\right) Re_{D1} Pr_i}{1 + 0.04 \left(\frac{D}{L_1}\right) Re_{D1} Pr_i}^{1/4} = 9.51 \neq 3.66$

$\bar{Nu}_{D1} = 16.9$ repeat for this val. $\approx 16.93 \approx 16.9 \checkmark \rightarrow 18.1 \text{ m}$

mean temp state 2 $\bar{T}_{m2} = \frac{52+80}{2} = 66^\circ\text{C} = 339\text{K}$

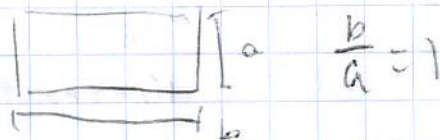
engine oil props outlet 339K from "Thermo Props" table

turb region Temp: $Re_{D2} = \frac{4\dot{m}}{\pi D \mu_2} = \frac{4}{0.005 \pi \cdot 5.62 \times 10^{-2}} = 4531.1$ $f = (0.79 \ln Re_{D2} - 1.64)^{-2} = 0.0398$

$\bar{Nu}_{D2} = \frac{(f/8)(Re_{D2}-1000)Pr_2}{1 + 12.7 \left(\frac{f}{8}\right)^{1/4} (Pr_2 - 1)} = 184$ $\bar{h}_2 = \frac{\bar{Nu}_{D2} k_2}{D_2} = 5120 \frac{\text{W}}{\text{m}^2\text{K}}$

$\frac{\ln(T_s - T_o)}{(T_s - T_{m1})} = \frac{-\pi D L_2 \bar{h}_2}{\dot{m} C_{p2}}$ $L_2 = \frac{\dot{m} C_{p2}}{\pi D \bar{h}_2} \ln\left(\frac{T_s - T_o}{T_s - T_{m1}}\right) = \frac{8.7 \text{ m}}{18.1}$
 $L_1 + L_2 = 26.8 \text{ m} \approx L$

8.66) Rect channel



for const Surface T.

$$Nu = 2.976$$

$$\therefore \frac{h D_h}{k} = 2.976 \quad h = \frac{2.976 k}{D_h}$$

$$D_h = \frac{4(a \times b)}{2(a+b)} = \frac{4a^2}{2(2a)}$$

$$D_h = a$$

$$D_h = \frac{4A}{P}$$

$$h = \frac{2.976 k}{a} \quad (1)$$

$$ii) \frac{b}{a} = 1.43 \text{ for } Nu = 3.183 = \frac{h D_h}{k}$$

$$D_h = \frac{4(a \times b)}{2(a+b)} = \frac{4(a \times 1.43)}{2(1+1.43)} \quad D_h = 1.176 a$$

$$h = \frac{3.183 k}{1.176 a} = 2.704 \frac{k}{a} \quad (2)$$

$$iii) \frac{b}{a} = 2 \quad Nu = 3.391 \quad D_h = \frac{4(a \times b)}{2(a+b)} = \frac{4(a \times 2a)}{2(a+2a)} \quad D_h = \frac{4}{3} a$$

$$\frac{h D_h}{k} = 3.391 \quad h = 3.391 \frac{k}{D_h} = 3.391 \times \frac{3}{4} a \frac{k}{a} \quad h = 2.543 \cdot \frac{k}{a} \rightarrow (3)$$

for $\frac{b}{a} = 1$ provide higher heat transfer than other condition.

$$\boxed{h_{\frac{b}{a}=1} > h_{\frac{b}{a}=1.43} > h_{\frac{b}{a}=2}$$

$$\text{for circular cross section } Nu = 3.657 \quad \frac{h D}{k} = 3.657 \quad h = 3.657 \frac{k}{a}$$

circ. tube heat transfer > non circular heat t.

$$h_{\text{circ}} > h_{\frac{b}{a}=1} > h_{\frac{b}{a}=1.43} > h_{\frac{b}{a}=2}$$

